MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 1102: INTRODUCTION TO PROGRAMMING WITH C

END OF SEMESTER EXAMINATION

DATE: SEPTEMBER, 2021 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are **17** questions on this paper. **ANSWER ALL.**

i. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Define the following terms:
   a. Function. **[1 Mark]**
   b. Object code. **[1 Mark]**
   c. Recursion. **[1 Mark]**
   d. Parameter. **[1 Mark]**
   e. Compiler. **[1 Mark]**

2. Given numbers from 1 to 30. Write a program in C that determines whether the number is even or odd. Implement using the following;
   a. **for** loop. **[3 Marks]**
   b. **while** loop. **[3 Marks]**
   c. **do while** loop. **[3 Marks]**

3. What is the difference between *call by value* and *call by reference*? **[4 Marks]**

4. Differentiate source code from object code. **[2 Marks]**

5. Write a program in C that solves a quadratic equation if the user inputs **a**, **b** and **c**. **[6 Marks]**

6. What is the advantage of an array over individual variables? **[2 Marks]**

7. Write an algorithm in a form of flow chart for a program that asks a user to enter a whole number. If the number is even, the program should calculate and print the cube of that number otherwise it should calculate and print the square of that number. **[4 Marks]**

8. What will be the output from the following program segment **(There are no errors)**: **[4 Marks]**

```
int m=3,n=6;
printf("The values are: ");
printf("%d\n ", ++n);
printf("%d\n ", n + - -m - ++n );
```

9. Write a program to calculate and print the salary of a new recruit as follows; if salary is below K900, tax should not be calculated otherwise a 0.8% tax should be deducted. The program should reject salary amount that is less than zero. **[5 Marks]**

10. A user wants to implement a function that will calculate bonus of a sales clerk who will get 10% of the total sales. Write a function prototype that will achieve this. The name of the function be ***BonusCalculator***. **[4 Marks]**

11. What is main the difference between a while loop and a do while loop? **[2 Marks]**

12. Use a for loop to write a program that calculates and prints a multiplication table as follows: *(Hint: Use nested for loops)* **[8 Marks]**

2	**4**	**6**	**8**	**10**	**12**	**14**	**16**	**18**
3	**6**	**9**	**12**	**15**	**18**	**21**	**24**	**27**
4	**8**	**12**	**16**	**20**	**24**	**28**	**32**	**36**

13. Explain the **two** arguments of a scanf function. **[4 Marks]**

14. Write a program that expects a certain number from a user. The user should keep entering the numbers until the intended number has been entered. When a user enters a number that is not required, the program should print a message "**Wrong Number. Try again!**", and allow the user to enter another number. Make an assumption on which number to use. **[5 Marks]**

15. Write a programme that calculates and prints the average of three decimal numbers entered by a user. **[4 Marks]**

16. Determine and correct the syntax errors from the following statements:

a. char letter = 2; **[2 Marks]**
b. float int; **[2 Marks]**
c. int X= 8; printf("%d", x); **[2 Marks]**
d. int y; for (y=0; y=10, --y) {printf("%d\n", y*y)} **[2 Marks]**
e. float tax = = 20%; **[2 Marks]**
f. char number = "21"; **[2 Marks]**

17. Write a program that gets a number from a user. The program should have a function called ***SquareRoot*** that should calculate and return the square root of the number. **[5 Marks]**

END OF QUESTION PAPER